

$$r^{(0)} = R_{g,N}$$

$$k_q = \operatorname{argmin}_k \|r^{(q-1)} - e_k^{(q)}\|_2^2$$

$$r^{(q)} = r^{(q-1)} - e_{k_q}^{(q)}, \quad q = 1, \dots, 4$$

$$\widetilde{X}_g = \widehat{X}_{g,N} + \sum_{q=1}^4 e_{k_q}^{(q)}$$